# Exit Ticket · reflect and retrieve

## Exit Ticket

If a player chooses an occupied square, the program should switch to the other player anyway.

## Use key vocabulary from memory

1. Explain 2D array in your own words.

2. Explain Row in your own words.

3. Explain Column in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 015 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Noughts & Crosses · Part 3 — Make moves

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which method from Part 2 displays the board?

### Q2

In a zero-indexed C# 2D array, which row and column does board[1,2] access?

### Q3

What do nested loops let us do with the board?

### Q4

Move 5 is the centre square. What zero-based row and column should it map to?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Map 1–9 into row/column — CODE:
int row = (move - 1) / 3;
int col = (move - 1) % 3; — Integer division finds the row; MOD finds the column. The picture highlights move 8: row 2, column 1. Inside MakeMove(), check the 1–9 range before using these indexes.

## Core idea 2

Check the move before accessing the array — CODE:

```
if (move < 1 || move > 9)
return false;

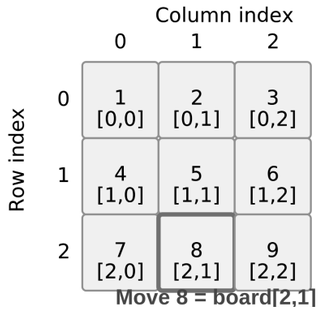
// Now calculate row and col.
// Only access the array for a valid index.
if (board[row,col] == "X" ||
board[row,col] == "O")
{
return false;
```

} — Reject choices outside 1–9 before calculating/using their indexes. Reject occupied squares too. Never overwrite an X or O.

## Core idea 3

Make the move — CODE:
board[row,col] = player;
return true; — A Boolean return value tells Main whether the move succeeded.

**Key visual · complete concept**








# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Make alternating turns work
Use your Part 2 DrawBoard() , prompt for a move, call MakeMove() , and switch player only after a valid move.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Map a move
Write code that converts an integer move from 1–9 into zero-based row and column values.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

Switch player only after success — CODE:
if (currentPlayer == "X")
currentPlayer = "O";
else
currentPlayer = "X"; — An invalid move should leave the same player to try again.

## Vocabulary · use the correct technical terms

2D array — An array organised using rows and columns.

Row — The horizontal index in a 2D array.

Column — The vertical index in a 2D array.

Nested loop — A loop inside another loop.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Turn move 8 into a board position

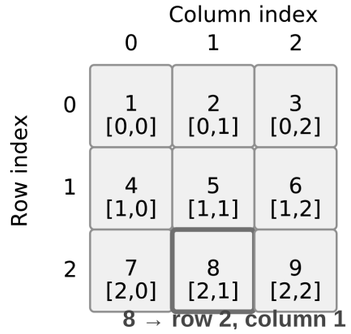
## Follow the worked method

1. Subtract 1 — CODE:
move - 1 = 7 — Board indexes begin at 0.

2. Use MOD 3 — CODE:
7 % 3 = 1
col = 1 — The remainder identifies the column.

3. Place the player — CODE:
board[2,1] = player; — Move 8 maps to row 2, column 1.

**Source model · final illustrated step**




8 → row 2, column 1

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Complete the same calculation
CODE:
row = (6 - 1) / 3 = ?
col = (6 - 1) % 3 = ?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.